Replication of Study Expecting the unexpected by Jiang & Filik (2025, Memory & Cognition)

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Introduction

Language comprehension requires readers to integrate incoming information with both real-world knowledge and the context established by a discourse. These sources of information can conflict when a narrative describes events that would be implausible in reality but are possible within an alternative world. Jiang and Filik (2025) investigated this interaction using dream contexts, which permit implausible events without explicitly predicting them. Their study

addresses whether real-world knowledge continues to guide comprehension in such contexts or whether readers adjust their interpretation to accommodate the alternative reality.

The present project will replicate Experiment 2 of Jiang and Filik (2025), which used self-paced reading to examine how this interaction unfolds during comprehension. Two competing predictions guide the replication: real-world knowledge dominance and contextual modulation. Under real-world knowledge dominance, implausible target words should elicit longer reading times than plausible target words in both realistic and unrealistic contexts. This plausibility effect may emerge at the critical word or in subsequent spillover regions and persist into later regions. Under contextual modulation, implausible target words should elicit longer reading times in realistic contexts, whereas this effect should weaken or reverse in unrealistic contexts as readers integrate the information with the dream context. A reversal would involve shorter reading times following implausible than plausible target words within unrealistic contexts.

This replication is closely related to my current research on how comprehenders use grammatical cues and world knowledge to construct mental representations of counterfactual sentences. Although dream narratives and counterfactual constructions establish alternative realities in different ways, both raise questions about how raeders accommodate information that conflicts with real-world knowledge. Replicating this experiment will therefore allow me to investigate a question relevant to my EEG research using a complementary behavioral measure of comprehension.

My research background is primarily qualitative and theory-focused and I began conducting experiments after starting my PhD approximately two years ago. This replication will provide valuable experience in experiment design, behavioral reading-time measurement, and mixed-effects modeling as I continue developing confidence in quantitative analysis. These skills will also support my broader research on the cognitive processes underlying language comprehension.

Methods

Power Analysis

I conducted simulation-based power analysis using the original data and fitted linear mixed-effects models with an unadjusted significance threshold of = 0.05. My primary interest is the main effect of plausibility at Spill3, where the original estimated effect was = 0.04649 on the log reading-time scale. Estimated power was 95% with 40 participants (95% Monte Carlo CI [88.72%, 98.36%]; 100 simulations), exceeding the 70% and 90% targets and reaching the 95% target in the point estimate. I also examined the plausibility effect at Spill2 (- 0.04954), where estimated power was lower. Based on 1000 simulations per sample size, estimated power was 82.5% with 48 participants, 84.9% with 52 participants, and approximately 89% with 60 participants. With 104 participants, estimated power was 96% although this estimate was based on only 100 simulations. These results suggest that Spill2 requires a larger sample than Spill3 to detect the original estimated effect.

Planned Sample

Balancing statistical power and recruitment feasibility, I provisionally plan to obtain 52 analyzable participants, allowing 13 participants per counterbalancing list. To obtain 52 analyzable participants, I plan to recruit approximately 64 participants, initially assigning 16 to each of the four counterbalancing lists. This recruitment target accounts for the original study's exclusion rate of 18.75%: 24 of 128 participants were excluded due to low comprehension accuracy or excessive outliers.

The original study recruited students from the University of Nottingham through the SONA system in exchange of course credit. The final sample included 104 participants (91 female; age: M = 18.87 years, SD = 0.68, range = 18-21 years). To recruit a comparable university student sample, I plan to recruit native English speakers through UCSD's SONA system in exchange of course credit.

Materials

The replication will use the materials from Experiment 2 of the original study, which crossed context (realistic vs. unrealistic) with real-world knowledge plausibility (plausible vs. implausible). Each critical scenario contained a critical word followed by a five-word spillover phrase. The authors described the spillover regions as follows: "The third part was a five-word spillover phrase (e.g., 'before she went back home'), which was always identical across all conditions of the same item, and was always in the same sentence as the critical word with no punctuation in between. This means that the Spill5 region was always the last word of the sentence."

Critical words were counterbalanced across scenarios so that the same word could serve as plausible or implausible information across participants, while each participant encountered that word only once. The authors described the item and list structure as follows: "Twenty-four sets of critical items were used in the experiment. Four lists were created using a Latin square procedure, with each list containing 24 critical items (six per condition) as well as 60 filler items. The 60 filler stimuli had a similar structure to the critical items, and also varied in length, but they were all realistic and did not contain any implausible information."

I plan to preserve this stimulus structure and counterbalancing scheme. However, I have located only 22 of the 60 filler items in the shared materials. I will contact the authors to request the remaining fillers and comprehension questions and will document any deviations if the complete set cannot be obtained.

Procedure

The experiment will be conducted online using PCIbex, the platform used in the original study. Although the authors shared stimulus lists and analysis scripts, I have not located the original experiment code. I will therefore recreate the task while preserving the original

stimuli, presentation format, response procedure, counterbalancing, and trial-order constraints as closely as possible.

After providing informed consent, "participants were instructed to read each scenario attentively and silently. Each trial began with an asterisk (*) in the middle of the screen. Participants pressed the spacebar when they were ready to read the scenarios. All the scenarios were presented word-by-word. The words initially appeared as a series of underlines obscuring the words (_____ _____ _____ _____), and participants pressed the spacebar to reveal each word. The presentation was noncumulative so previous words were replaced with underlines when the next word appeared, which means participants could only see one word at a time. A third of the materials were followed by a statement that probed comprehension. Participants were required to judge whether the statement was true or false based on the previously read scenario by pressing one of two keys. The assignment of left/right hands to true/false responses was completely random. Reading times for each word and the accuracy of participants' responses to the comprehension questions were recorded."

The procedure described above will be followed in the replication.

Analysis Plan

Based on the original findings, the primary outcome will be the main effect of plausibility at the third spillover region (Spill3) with the corresponding effect at Spill2 examined as an additional outcome. To assess contextual modulation, the context x plausibility interaction at Spill5 will be examined as a secondary outcome and its estimate and confidence interval will be reported. Spill5 is always the sentence-final word, where sentence wrap-up processes may encourage readers to integrate the sentence with the broader discourse context. These outcomes distinguish the early influence of real-world plausibility from later changes in its influence across contexts.

For the analysis, I will follow the original study by starting with the maximal random-effects structure. If convergence issues arise, I will simplify the structure using a prespecified sequence and document any changes.

Data Preparation and Exclusion Rate

Following the original study, I will exclude participants whose overall accuracy on comprehension question is below 80% or whose proportion of outlier reading time exceeds 30%. Based on the original study's overall exclusion rate, I expect to have 52 participants as my final sample.

Data analysis will be performed over six regions of interest separately: one critical region and five spillover regions. First, I will remove the response times for the comprehension questions below 100 ms or above 5,000 ms as 100 ms is too short to make the decision for the question and 5,000 ms is considered too long than required. For the reading times, I will prune those

that are over 2.5 SDs from the grand mean. Data within the range will be included in the analysis, regardless of the accuracy on comprehension questions.

Differences from Original Study

Several differences between the original study and the planned replication should be acknowledged. First, the original study recruited students from a UK university, whereas the replication will recruit native English-speaking students from UC San Diego. Differences in familiarity with British and American English vocabularly, expressions, or cultural differences could influence reading times. The original study did not directly test whether plausibility effect varies across these populations, so the impact of this difference remains uncertain. I will review the materials for potentially unfamiliar language and document any adaptations while preserving the original manipulation as closely as possible.

Second, the planned sample will be smaller. The original study analyzed data from 104 participants, whereas I plan to recruit approximately 64 participants to obtain an estimated 52 analyzable participants after exclusions. This decision reflects recruiment feasibility and the simulation-based power analyses. A smaller sample will reduce precision and power, particularly if the original effect size overestimate those in the replication.

Last, the complete set of filler items and comprehension questions is not currently available to me. I will contact the authors to request the remaining materials, however, if these cannot be obtained, replacement materials will be needed, and all deviations will be documented. Changes to the fillers could alter the proportion of implausible events and readers' expectations, while changes to comprehension questions could affect task demands, accuracy, and participants exclusions.

Methods Addendum (Post Data Collection)

You can comment this section out prior to final report with data collection.

Actual Sample

Sample size, demographics, data exclusions based on rules spelled out in analysis plan

Differences from pre-data collection methods plan

Any differences from what was described as the original plan, or "none".

Results

Data preparation

Data preparation following the analysis plan.

Confirmatory analysis

The analyses as specified in the analysis plan.

Side-by-side graph with original graph is ideal here

Exploratory analyses

Any follow-up analyses desired (not required).

Discussion

Summary of Replication Attempt

Open the discussion section with a paragraph summarizing the primary result from the confirmatory analysis and the assessment of whether it replicated, partially replicated, or failed to replicate the original result.

Commentary

Add open-ended commentary (if any) reflecting (a) insights from follow-up exploratory analysis, (b) assessment of the meaning of the replication (or not) - e.g., for a failure to replicate, are the differences between original and present study ones that definitely, plausibly, or are unlikely to have been moderators of the result, and (c) discussion of any objections or challenges raised by the current and original authors about the replication attempt. None of these need to be long.